

EXPERIMENTAL FACT

OUR VERDICT

WHY IT FOLLOWS

Anhydrous (5.5 Å) is not SC

Foo 2003



explained

$\alpha > 0$: single stiff well ($\hbar\omega \approx 30$ meV),
 $N(0) \approx 0$ — no 2DEG, nothing to pair.

Monolayer hydrate (6.9 Å) is not SC

Foo 2003



explained

Past the window: $\lambda \approx 43 \gg 2$ — the alkali
self-traps (polaronic), so SC is killed.

Bilayer hydrate (9.9 Å) is SC, $T_c = 4.5$ K

Takada 2003



prediction

Bare geometry is polaronic here; SC needs water to
soften the Na well back into the window (untested).

No superconducting Li analogue



explained

$c_{\text{Li}}^* \leq 5.5$ Å and lighter mass $\Rightarrow$ stiffer mode;
Li never enters the soft-mode window.

Magnetic phase borders the dome

expt.



explained

Local moment turns on exactly at the $\alpha < 0$ well
transition (Fig. 4), coincident with the dome.

Superconducting T_c is only a few K

Takada 2003



explained

Allen–Dynes with DFT ω_{eff} , λ gives peak
 $T_c \approx 14$ K (band 11–23) — the right scale.